



Phishing Between Loneliness and Skepticism: How Social Engineering Exploits the Need for Belonging, Safety, and Control

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Student number: [fill in]

Master's Thesis, Phase 1 (ACIT5910)

Long master's thesis (60 ECTS)

OsloMet - Oslo Metropolitan University

Faculty of Technology, Art and Design

Master's Programme in Applied Computer and Information Technology



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Abstract

Phishing remains one of the most common entry points to serious cyber incidents, despite rising public awareness and technical controls. This thesis investigates why phishing continues to work in a context characterised by both increased loneliness and increased general skepticism toward strangers online. The core argument is that phishing often does not resemble a stranger attack. Instead, it imitates what people still long for or fear losing: belonging, safety, order, status, help, and control.

The thesis introduces the **skepticism paradox**: general caution toward unknown senders may coexist with heightened openness to messages that feel socially or institutionally legitimate. Using exploratory sequential mixed methods among adults in higher education in Norway, the study examines emotional and contextual vulnerability markers, the gap between knowing about phishing and acting safely under pressure, and which protective factors may reduce risk in practice.

The expected contribution is threefold: (1) a human-centred model of phishing susceptibility that moves beyond the "weakest link" narrative; (2) empirical insight into how loneliness and skepticism interact in Norwegian higher education; and (3) practical recommendations linking student welfare, security culture, and awareness design at OsloMet and comparable institutions.

Keywords: phishing, loneliness, social engineering, human factors, protection motivation, cybersecurity behaviour

1. Introduction

Cybersecurity is often presented as a technical problem with a human weak point. This thesis starts from a different premise: phishing is a human decision problem disguised as an email.

Every year, organisations invest in filters, training, policies, and incident response. Yet phishing and related social engineering attacks remain among the most effective ways to gain initial access to systems, accounts, and sensitive information (ENISA, 2024; CISA, n.d.). The problem persists not only because attackers improve their tools, but because they improve their understanding of human motivation.

In Norway, public awareness of digital fraud has increased. People are told to be skeptical of unknown senders, unexpected links, and requests for credentials. At the same time, loneliness and social disconnection are widely discussed as social problems across age groups (Statistics Norway, 2025; World Health Organization, 2023). This creates a tension that is central to the thesis:

In a time of both rising loneliness and rising general skepticism, phishing becomes dangerous precisely because it does not look like an attack from a stranger. It imitates what people still long for or fear losing: belonging, safety, order, status, help, and control.

The thesis therefore asks not only why people click, but why emotionally and socially meaningful lures can bypass skepticism that would otherwise protect them from obvious fraud.

1.1 Background and relevance

Recent Norwegian warnings from Finans Norge, banks, NSM, and Nkom show that social engineering remains a live threat. Attacks combine email, SMS, phone calls, impersonation, urgency, and authority. AI-assisted phishing may increase credibility and scale, but the underlying mechanism remains psychological: influence under uncertainty (Halcyon, 2026; Microsoft, 2024; Norwegian National Security Authority, 2026).

For a Norwegian master thesis in cybersecurity, this topic is timely and feasible. It connects technical security with human behaviour, privacy, trust, and social context - without requiring access to classified infrastructure data.

1.2 Research problem

Main research question:

How can phishing remain effective in a context of increased loneliness and increased general skepticism, and which emotional, cognitive, and contextual factors make individuals especially vulnerable to this form of social engineering?

Sub-questions:

1. How do loneliness, social connection needs, and general online skepticism relate to phishing susceptibility?
2. Which emotional and situational factors (urgency, authority, fear, shame, stress) most strongly influence unsafe responses?
3. What is the gap between knowing about phishing and acting safely in the moment of decision?
4. Which protective factors - training, culture, verification routines, peer support - are experienced as most effective?

1.2.1 The skepticism paradox

This thesis names and examines a specific tension: the **skepticism paradox**. People in Norway are repeatedly told to distrust unknown links, unexpected requests, and unfamiliar senders (Finans Norge, 2026; Nettvett.no & Norwegian National Security Authority, n.d.). At the same time, loneliness and social disconnection are elevated among young adults and students (Statistics Norway, 2025; Khrono & Norwegian Student Organisation, 2024). Skepticism may therefore protect against obvious stranger-framed fraud while leaving individuals exposed to lures that imitate belonging, institutional authority, or timely help.

If training assumes that skepticism is uniformly protective, organisations may overlook the emotional and social conditions under which credible impersonation succeeds. If loneliness is treated only as a welfare issue, institutions may miss its relevance to fraud susceptibility.

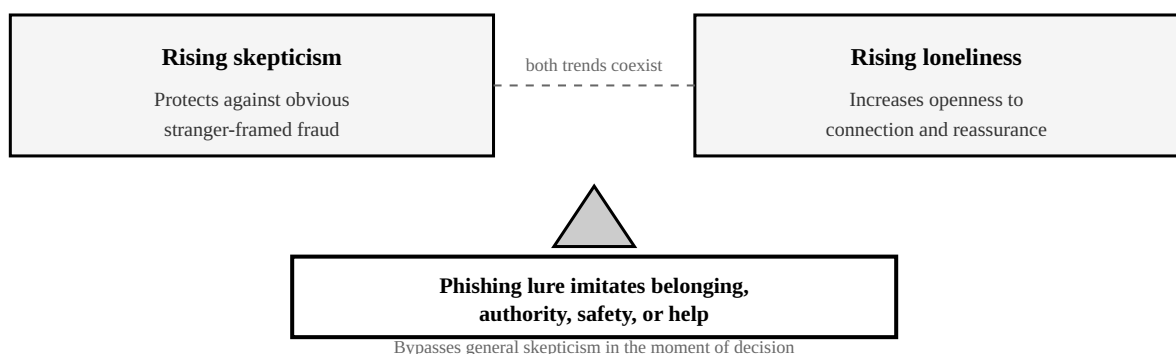


Figure 2. The skepticism paradox: two simultaneous social trends create a vulnerability window.

Source: Original model for this thesis.

1.3 Scope

- **In scope:** phishing, smishing, vishing, impersonation scams, emotional and social manipulation online.
- **Population:** adults in higher education in Norway, primarily 18-35 years.
- **Out of scope as main focus:** malware reverse engineering, DDoS, ransomware operations, deep technical forensics.
- **Geographic frame:** Norway, with international literature as theoretical backdrop.

1.4 Contribution

The thesis aims to contribute academically and practically:

- **Theoretical:** a combined model of phishing vulnerability integrating loneliness, skepticism, situational pressure, and security culture - extending Protection Motivation Theory where threats are not initially recognised as such (Rogers, 1975; Floyd et al., 2000).
- **Empirical:** mixed-methods evidence from Norwegian higher education on the skepticism paradox, using validated loneliness measures, scenario vignettes, and semi-structured interviews (Braun & Clarke, 2006).
- **Practical:** recommendations for awareness, reporting culture, and student welfare integration at OsloMet and comparable institutions - linking technical security with human meaning rather than user blame (Hadnagy, 2018; Chen et al., 2024).

Norwegian studies by Tjostheim and Waterworth (2020, 2022) show that cognitive style predicts phishing susceptibility in representative samples. This thesis examines whether *emotional and social* variables add explanatory power beyond cognitive style alone.

1.5 Thesis structure

This Phase 1 submission (ACIT5910) comprises the introduction, background, literature review, theoretical framework, and methodology. Chapters 6-7 outline the planned analysis and discussion for Phase 2 (ACIT5920) and Phase 3 (ACIT5930). Chapter 8 presents preliminary conclusions and recommendations. Appendices contain the interview guide, survey instrument, scenario vignettes, ethics application, timeline, and consent materials.

2. Background: Phishing as an Entry Point

2.1 What is phishing?

Phishing is a social engineering attack in which the victim is manipulated into clicking a malicious link, opening an attachment, revealing credentials, approving an action, or transferring money. The attack succeeds when the message appears credible, relevant, or emotionally meaningful (Hadnagy, 2018; CISA, n.d.).

2.2 Phishing as the starting point in attack chains

Many serious incidents begin with a human action: a click, a reply, a login, or an approval. From there, attackers may pursue account takeover, data theft, business email compromise, or further exploitation. This makes phishing analytically important even when the final harm is ransomware or fraud.

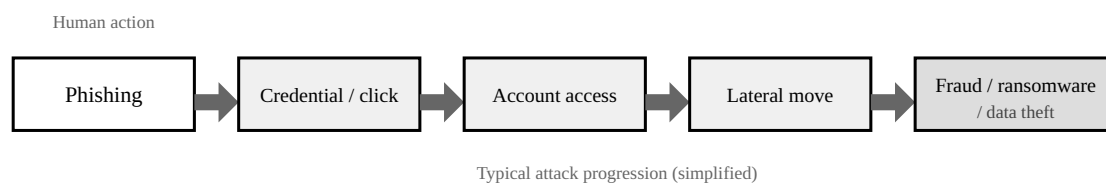


Figure 1. Phishing as the human entry point in a typical cyber attack chain.

Source: Adapted from ENISA (2024) and CISA (n.d.).

2.3 Norway and digital trust

Norway is highly digitalised. Citizens rely on BankID, ID-porten, Altinn, digital mailboxes, and online banking. High trust in institutions can be a societal strength, but it may also create fertile ground for impersonation attacks when combined with urgency and fear. Public campaigns such as Finans Norge's "self-defence against fraud" (selvforsvar mot svindel) show that awareness exists - yet incidents continue.

2.4 AI and the new scale

AI can improve grammar, personalisation, translation, and impersonation. That lowers the cost of credible phishing, but it does not replace the need for emotional leverage. AI makes the lure easier to produce; human needs determine whether it works.

2.5 Norwegian context: loneliness, students, and phishing

Norway combines high digital trust with measurable social disconnection in key groups. SSB's Livskvalitetsundersøkelse (2025) finds that 14.5% of adults report being quite or very troubled

by loneliness in the last two weeks, with the highest rates among 18-24-year-olds (23.3%) and 25-44-year-olds (19.1%) (Statistics Norway, 2025). Among students specifically, a 2024 NSO/Khrono survey of 1,000 students found that 68% report some degree of loneliness, including 22% who feel lonely to a large or very large extent (Khrono & Norwegian Student Organisation, 2024).

At the same time, phishing remains the dominant form of digital fraud in Norway. NSM received more than 100,000 reports of phishing attempts in 2024, and the 2026 risk report emphasises that social manipulation attacks are increasing in sophistication, including AI-assisted content and attacks that bypass traditional MFA (Norwegian National Security Authority, 2026). Norwegian population studies also show that intuitive thinking style and willingness to share personal data predict phishing susceptibility in national samples (Tjostheim & Waterworth, 2020, 2022). This juxtaposition - high loneliness among young adults and students, plus rising phishing volume - makes the thesis question locally relevant, not only theoretically interesting.

Table 1. Norwegian context indicators: loneliness and phishing threat. *Sources: Statistics Norway (2025); Khrono & Norwegian Student Organisation (2024); Norwegian National Security Authority (2026).*

Indicator	Source	Relevance to thesis
14.5% adults quite/very lonely (2025)	SSB Livskvalitet	Baseline population context
23.3% of 18-24 report strong loneliness	SSB	Core age group in HE sample
68% students report some loneliness	NSO/Khrono 2024	OsloMet/HE recruitment frame
100,000+ phishing reports (2024)	NSM	Threat is live and measurable
AI-assisted phishing rising	NSM Risiko 2026	Human factors remain decisive

3. Literature Review

3.1 Phishing and human factors

Research consistently shows that phishing success depends on persuasion strategies, context, workload, and decision pressure rather than intelligence alone (Parker & Flowerday, 2020; Tsai et al., 2021). Employees and students may know the rules yet still click when messages appear legitimate, urgent, or socially relevant.

3.2 Loneliness, belonging, and fraud

Loneliness is not merely the absence of company. Cacioppo and Patrick (2008) describe it as a distress signal indicating unmet social connection needs - a state that can increase openness to contact, reassurance, and perceived opportunity. The World Health Organization (2023) now treats social connection and loneliness as global health priorities, underscoring that disconnection is a structural rather than marginal concern.

Loneliness has long been associated with fraud vulnerability among older adults (Alves & Wilson, 2008; Wen et al., 2022). Recent experimental work suggests loneliness can increase susceptibility to misleading offers in middle-aged and older adults (Wen et al., 2024). Mechanisms include increased desire for connection, greater trust in persuasive strangers, and reduced access to social verification.

However, the relationship is not uniform across age groups. Some studies find weaker or more complex effects among younger adults, where FOMO, social media exposure, impulsivity, and platform-specific cues may matter more (Przybylski et al., 2013; Klütsch et al., 2024). Among college students, Du et al. (2025) found that loneliness and impulsivity chain-mediate the relationship between cognitive failure and gullibility - suggesting that social and emotional pathways may be as important as cognitive style in fraud susceptibility.

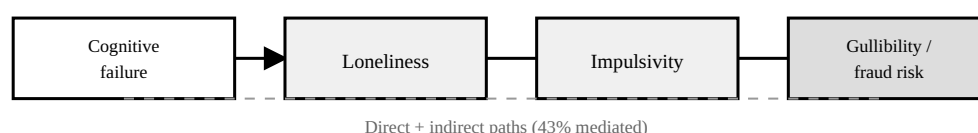


Figure 4. Chain mediation from cognitive failure to gullibility via loneliness and impulsivity (college students).

Source: Adapted from Du et al. (2025, Figure 1).

Research gap: loneliness is well studied in older fraud victims, but less systematically examined in Norwegian higher-education contexts and in relation to the skepticism

paradox.

At OsloMet and across Norwegian higher education, student loneliness is not marginal. SHoT 2022 found that 29% of students often miss someone to be with, and campus presence correlates with lower loneliness (SHoT Study, 2022). The Norwegian Directorate of Health (2024) and student welfare organisations such as SiO and Samskipnaden treat loneliness as a public-health and student-life issue (Norwegian Directorate of Health, 2024; Studentsamskipnaden i Oslo, 2024; Samskipnaden, 2025). OsloMet initiatives such as the Lunch Friend (Lunsjvenn) project confirm that institutions recognise social belonging as a welfare concern (OsloMet, 2024). This thesis connects that welfare concern to cybersecurity: the same need for connection that drives welfare interventions may also shape responses to socially framed phishing.

3.3 The skepticism paradox in practice

People may report high skepticism toward unknown senders yet remain vulnerable to messages that appear official, socially relevant, or emotionally important. Young adults are highly exposed on social platforms and may respond differently to phishing that mimics peers, events, or opportunities rather than strangers (Klüttsch et al., 2024; Parker & Flowerday, 2020). Agrawal et al. (2025) found that Gen Z participants often recognise security risks yet remain uncertain about how to respond and seek support - a pattern consistent with awareness without behavioural confidence.

This is the empirical face of the skepticism paradox: knowing that fraud exists does not guarantee safe action when a message feels meaningful, urgent, or socially rewarding. Technical indicators (misspelled domains, odd links) may be less salient than emotional relevance in the moment of decision (Hadnagy, 2018; Conteh & Schmick, 2020).

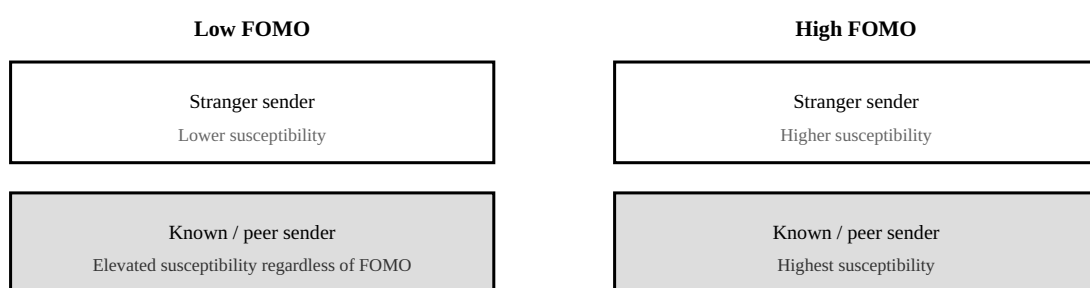


Figure 5. Known senders and fear of missing out (FOMO) increase phishing susceptibility on social media.

Source: Adapted from Klüttsch et al. (2024).

3.4 Protection Motivation Theory (PMT)

PMT explains protective behaviour through threat appraisal (severity and vulnerability) and coping appraisal (response efficacy and self-efficacy) (Rogers, 1975; Floyd et al., 2000). In

cybersecurity, PMT has been used to explain reporting behaviour and resistance to phishing (Tsai et al., 2021; Bayl-Smith et al., 2024).

For this thesis, PMT is useful but incomplete. It explains motivation to protect oneself, yet phishing often succeeds when the message does not initially look like a threat. Emotional needs may reframe the lure as an opportunity, warning, or social contact rather than danger.

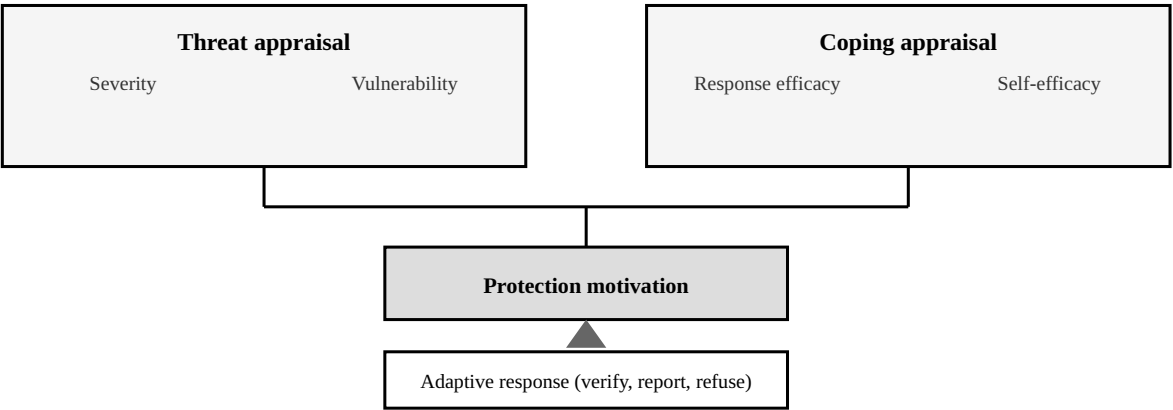


Figure 3. Protection Motivation Theory: threat and coping appraisals combine to produce protective behaviour.

Source: Adapted from Rogers (1975) and Floyd et al. (2000).

3.5 Remote work, cognitive bias, and shame

Remote and screen-mediated communication reduces social cues available for verification. Social engineering research shows that isolation, trust exploitation, and reduced oversight can increase susceptibility to manipulation (Conteh & Schmick, 2020). Overconfidence bias may coexist with loneliness: people may believe they are "too smart" for phishing while remaining vulnerable to emotionally relevant lures (Parker & Flowerday, 2020).

Shame and fear of judgment reduce reporting. Training that relies on simulated phishing without support-seeking components may increase awareness but not safe behaviour under stress (Chen et al., 2024). For lonely individuals, romance and confidence scams show that victims may return to contact with scammers because the relationship itself feels meaningful (WIRED, 2024). Universities need non-punitive reporting cultures if data collection and prevention are to reflect real behaviour.

3.6 Theoretical synthesis and original contribution

Table 2. Theoretical perspectives synthesised in the literature review.

Perspective	Contributes	Limitation
-------------	-------------	------------

Perspective	Contributes	Limitation
Social engineering / persuasion	Authority, urgency, fear, liking, scarcity	Can underplay emotional loneliness
Loneliness research	Connection needs, social void, trust in strangers	Often focused on older adults
PMT	Threat and coping appraisals	Assumes threat is recognised as such
Security culture	Reporting, verification, non-punitive learning	Needs organisational context

The thesis proposes a combined model centred on the **skepticism paradox: phishing susceptibility = emotional relevance of lure + situational pressure + unmet social need - verification opportunity - non-punitive security culture**. General skepticism moderates this relationship primarily for stranger-framed lures; it is weaker when the message imitates institutions, peers, or opportunities that feel legitimate.

Norwegian national studies complement international findings: Tjostheim and Waterworth (2020, 2022) demonstrate that cognitive reflection and data-sharing willingness predict phishing compliance in representative Norwegian samples. This thesis extends that line by asking whether *emotional* and *social* variables - especially loneliness and the need for belonging - add explanatory power beyond cognitive style alone, and whether the skepticism paradox helps explain why awareness campaigns have not eliminated phishing in a high-trust, high-loneliness context.

4. Theoretical Framework and Hypotheses

4.1 The skepticism paradox as analytical lens

The theoretical contribution of this thesis is to treat phishing not primarily as a failure of knowledge, but as a collision between two simultaneous social trends: rising general skepticism toward strangers and rising subjective loneliness or social need (Statistics Norway, 2025; Khrono & Norwegian Student Organisation, 2024). The **skepticism paradox** predicts that protective skepticism is lure-dependent: it works against obvious stranger fraud but weakens when messages imitate belonging, authority, order, status, or help.

Phishing succeeds when it imitates legitimate human and institutional needs more convincingly than general skepticism can counter in the moment of decision.

4.2 What phishing imitates

Table 3. Human needs that phishing imitates and why general skepticism may fail. *Adapted from Hadnagy (2018) and thesis model.*

Need / fear	Example lure	Why it may bypass skepticism
Belonging	Message from "colleague", romance scam, event invite	Feels socially relevant, not random
Safety	"Your account is compromised"	Triggers protective action
Order / control	Tax, invoice, admin deadline	Promises to fix a problem quickly
Status	Job offer, exclusive opportunity	Appeals to identity and ambition
Help	IT support, "we can fix this for you"	Uses helpfulness and compliance

4.3 Hypotheses

1. **H1:** Higher loneliness is associated with higher self-reported phishing susceptibility, especially for socially framed lures (S2, S6).
2. **H2:** Higher general skepticism reduces susceptibility for stranger-framed lures but not for institutionally or emotionally relevant lures (interaction effect).
3. **H3:** Situational pressure (urgency, authority, stress) partially mediates the relationship between loneliness and unsafe intended behaviour.

4. **H4:** Participants with stronger verification habits and supportive, non-punitive security culture report lower susceptibility (PMT coping appraisal).

Age will be treated as a control/exploratory variable because prior research suggests loneliness may operate differently across cohorts (Wen et al., 2024).

4.4 Conceptual model

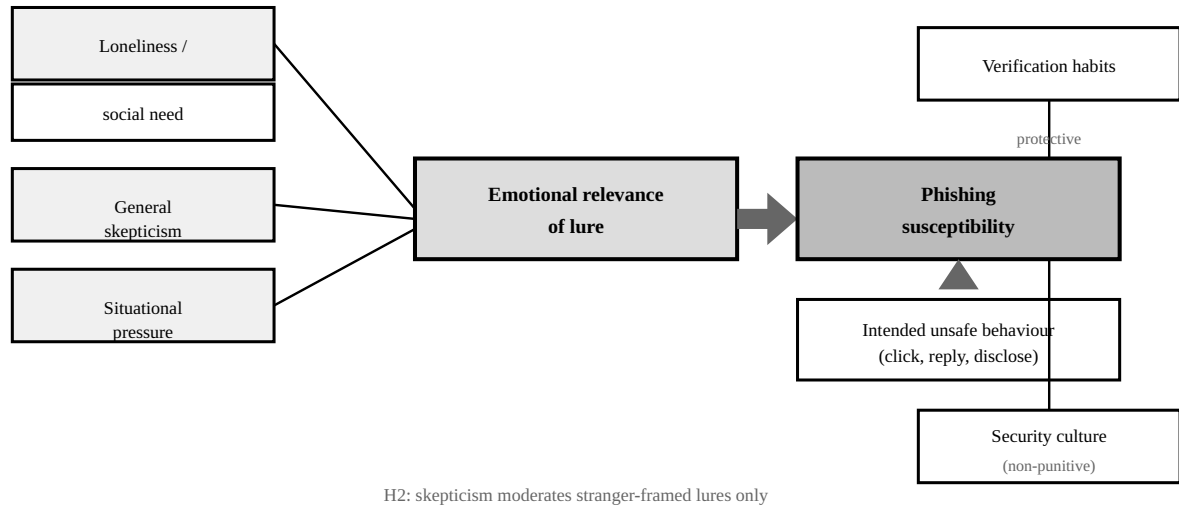


Figure 6. Combined conceptual model: the skepticism paradox and protective factors.

Source: Original model for this thesis, integrating PMT (Rogers, 1975) and loneliness research (Cacioppo & Patrick, 2008; Du et al., 2025).

The model predicts that skepticism reduces susceptibility primarily for stranger-framed lures, while loneliness and situational pressure increase susceptibility for belonging- and authority-framed lures. Protective factors operate both directly and by increasing opportunity for verification before action.

5. Methodology

5.1 Design

Recommended design: exploratory sequential mixed methods.

1. Phase 1: Online survey (quantitative + short open text)
2. Phase 2: Semi-structured interviews with selected respondents
3. Phase 3: Thematic analysis integrated with survey findings

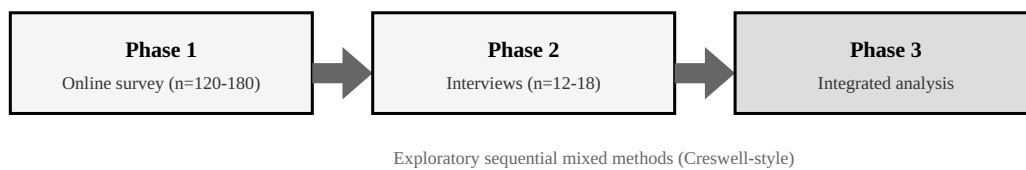


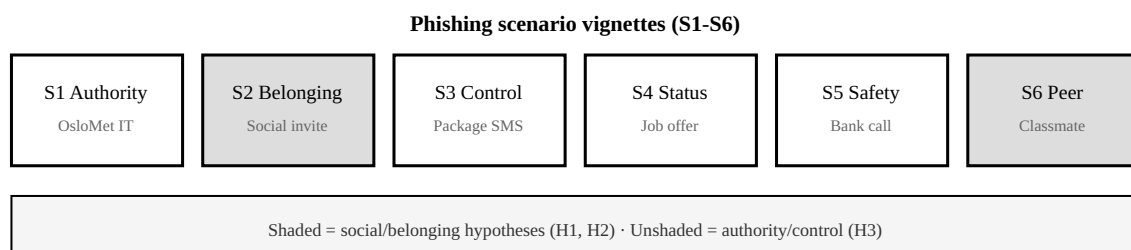
Figure 7. Three-phase exploratory sequential mixed methods design.

Source: Original design for this thesis; qualitative integration follows Braun and Clarke (2006).

5.2 Population and sample

- **Population:** students and staff in higher education in Norway.
- **Target sample:** 120-180 survey responses; 12-18 interviews.
- **Inclusion:** adults 18+ with regular digital communication for study/work.
- **Recruitment:** OsloMet programmes, student networks, LinkedIn, privacy-preserving invitation text.

5.3 Survey instruments



Full vignette text in Appendix C · Based on Hadnagy (2018) lure taxonomy

Figure 8. Overview of six scenario vignettes mapped to lure types and hypotheses.

Source: Original instrument for this thesis; lure categories adapted from Hadnagy (2018).

Table 4. Survey constructs, instruments, and validation notes.

Construct	Instrument / measure	Notes
Loneliness	UCLA Loneliness Scale (short form) or comparable validated scale	Must be licensed/validated for Norwegian use
General skepticism	Adapted items on trust in unknown senders / online caution	New scale, pilot first
Phishing susceptibility	Scenario-based items + self-reported past experience	No real credential capture
Situational factors	Stress, workload, urgency sensitivity	Likert items
Protective behaviour	Verification habits, reporting, training recall	PMT-informed
Demographics	Age, study field, digital confidence	Controls

5.4 Interview guide (summary)

Interviews will explore:

- near misses or actual encounters with phishing;
- what the message looked like and why it felt credible;
- emotional state at the time (lonely, stressed, busy, afraid);
- whether they knew the advice but still hesitated or clicked;
- what would have helped them stop.

Full guide in Appendix A.

5.5 Analysis plan

- **Quantitative:** descriptive statistics, correlation, regression, mediation tests where assumptions hold.
- **Qualitative:** reflexive thematic analysis (Braun & Clarke, 2006).
- **Integration:** joint display of statistical patterns and lived experience quotes.

5.6 Ethics

- Application to NSD / institution ethics board before data collection.
- Informed consent, right to withdraw, anonymisation.
- Special care when asking about fraud victimisation or shame.
- No deceptive live phishing against participants without explicit ethical approval.

5.7 Limitations (anticipated)

- Self-report bias and social desirability.
- Convenience sample may overrepresent cybersecurity-aware students.
- Loneliness scales measure subjective experience, not objective isolation.
- Causality cannot be fully established in cross-sectional survey data.
- Wen et al. (2024) found weaker loneliness-fraud effects in younger adults; H1 may need nuance or subgroup analysis.

5.8 Pilot study plan

Before main data collection, a pilot with 10-15 OsloMet students/staff will test: (a) completion time (target under 12 minutes), (b) clarity of scenario vignettes, (c) floor/ceiling effects on loneliness items, and (d) whether skepticism items discriminate. Pilot feedback will refine wording and interview prompts.

5.9 Data management

- Survey hosted on institution-approved platform (e.g. Nettskjema / Qualtrics).
- Data stored encrypted on OsloMet-approved storage; access limited to researcher and supervisor.
- Retention aligned with NSD approval; deletion after thesis archive period.
- Interview audio transcribed; direct identifiers removed before analysis.

6. Planned Analysis (Phase 2 and 3)

6.1 Descriptive findings (planned)

- Sample profile: age, role, study field, digital confidence, loneliness distribution.
- Self-reported phishing exposure: frequency, channels (email, SMS, social media, phone).
- Mean susceptibility scores by scenario type (S1-S6).
- Reported awareness vs. reported safe behaviour gap.

6.2 Hypothesis testing (planned)

Table 5. Planned hypothesis tests and expected directions (to be completed after data collection).

Hypothesis	Analysis	Expected direction
H1: Loneliness → susceptibility	Correlation / regression; compare S2 vs S1	Positive for social lures
H2: Skepticism alone insufficient	Interaction: skepticism × lure type	Skepticism weaker for institutional lures
H3: Situational pressure mediates	Mediation analysis (loneliness → stress → susceptibility)	Partial mediation
H4: Protective factors reduce risk	Regression with verification habits, reporting culture	Negative association

6.3 Qualitative themes (planned)

Interview analysis will use reflexive thematic analysis (Braun & Clarke, 2006) to identify themes such as:

- **Knowing but not stopping** - awareness without behavioural control under pressure.
- **Social credibility** - when messages feel like connection rather than threat.
- **Shame and silence** - barriers to reporting near misses.
- **What would help** - verification routines, peer support, non-punitive culture.

7. Planned Discussion (Phase 2 and 3)

7.1 Expected findings

The thesis may find that:

- participants report high awareness but uneven safe behaviour;
- loneliness and stress increase openness to socially meaningful lures;
- skepticism toward strangers does not protect against authority or belonging cues;
- verification culture and non-shaming reporting matter as much as training content.

7.2 Implications for theory

If supported, the findings would suggest that Protection Motivation Theory alone is insufficient for phishing because many lures are not initially appraised as threats (Rogers, 1975; Bayl-Smith et al., 2024). The skepticism paradox offers a human-centred explanation: protective motivation depends on whether the message is framed as danger or as opportunity, contact, or institutional routine. A combined model integrating emotional relevance, loneliness, verification habits, and security culture may better explain why skeptical individuals still fall for credible impersonation.

7.3 Practical implications

- Awareness campaigns should address emotional manipulation, not only fake links.
- Universities should normalise "pause and verify" without shame.
- Security training should include stress, loneliness, and social pressure scenarios.
- Design and process should make the safe action easier than the risky one.

7.4 Limitations

- Scenario-based susceptibility measures intended behaviour, not actual clicks.
- Convenience sample may limit generalisation beyond Norwegian HE.
- Cross-sectional design limits causal claims.
- Self-report of loneliness and victimisation subject to social desirability.

8. Conclusion and Recommendations

8.1 Conclusion direction

Phishing is best understood as an attack on human meaning, not merely human ignorance. In a more skeptical yet often lonelier digital environment, defenders must protect not only systems but the moments when people seek connection, certainty, and control. The skepticism paradox implies that cybersecurity education must address emotionally credible lures - not only technical indicators - and that student welfare and security culture are linked concerns rather than separate administrative domains.

8.2 Recommendations for OsloMet and higher education

1. **Integrate emotional manipulation** into security awareness - not only technical indicators.
2. **Normalise near-miss reporting** without academic or disciplinary penalty.
3. **Link student welfare and security** - loneliness programmes (e.g. Lunch Friend) and IT security are related.
4. **Scenario-based training** using HE-relevant vignettes (OsloMet IT, exam deadlines, peer messages).
5. **Verify-by-default routines** - separate channel verification for unexpected requests.

8.3 Future research

- Longitudinal study of loneliness and phishing incidents over an academic year.
- Intervention study: emotional-aware training vs. standard awareness.
- Comparison across Nordic HE institutions.

Appendix A: Interview Guide

1. Can you tell me about a time you received a suspicious message online - whether you clicked or not?
2. What made it seem believable or urgent?
3. How were you feeling that day (busy, stressed, lonely, distracted)?
4. Did the message feel like it came from a stranger, an institution, or someone you knew?
5. Did you know the security advice in that moment? If yes, why was it hard to follow?
6. Did you tell anyone? Why or why not?
7. What would help you stop faster next time?
8. Do you think loneliness or social pressure online affects how people respond to such messages?

Appendix B: Survey Questions

Section 1 - Background

- Age group
- Role (student / employee / other)
- Hours per day online for study/work
- Self-rated digital confidence (1-7)

Section 2 - Loneliness and social context

- UCLA-3 or equivalent validated items
- "I often feel that I lack companionship" (Likert)
- "I feel in tune with the people around me" (reverse)

Section 3 - Skepticism and awareness

- "I am generally skeptical of messages from unknown senders."
- "I feel confident that I can recognise phishing."
- "I have received cybersecurity training in the last 12 months."

Section 4 - Scenario susceptibility

Present 6 short vignettes. For each: "How likely would you be to click, reply, or follow instructions?" (1-7 Likert). See Appendix C for full vignette text.

Section 5 - Protective behaviour

- "I verify unexpected requests through a separate channel."
- "I would report a suspicious message even if I almost fell for it."
- "I worry about being judged if I report a near miss."

Appendix C: Phishing Scenarios

Each vignette is followed by a 1-7 likelihood scale and optional open comment. Scenarios are fictional and contain no live links.

Table 6. Full scenario vignettes for survey instrument.

#	Lure type	Scenario text	Construct tested
S1	Authority / safety	You receive an email from "OsloMet IT Security" saying your student account will be locked in 2 hours unless you verify your password via a link. You are preparing for an exam submission deadline today.	Urgency + institutional trust
S2	Belonging / social	A person you met briefly at a student event messages you on Instagram: "Hey, we should grab coffee - here's a link to my calendar." You have few friends on campus and have felt lonely lately.	Social relevance + loneliness
S3	Order / control	An SMS says a package is waiting and you must pay a 39 NOK customs fee via a link to avoid return. You recently ordered something online but are not sure which store.	Uncertainty + small cost
S4	Status / opportunity	LinkedIn message: "We saw your profile and want to invite you to a paid research assistant role. Complete this short form today - limited spots." You are job hunting.	Ambition + scarcity
S5	Help / compliance	Phone call from someone claiming to be bank support: "We detected suspicious activity. Stay on the line and approve the notification on your phone so we can secure your account."	Authority + fear
S6	Peer impersonation	Email from a classmate's address: "Can you quickly buy these gift cards for our group project? I'll pay you back tonight." The tone matches how they usually write.	Known sender + helpfulness

Analysis note: S2 and S6 test social/belonging hypotheses; S1, S3, S5 test authority and control; S4 tests status. Compare responses by loneliness quartile and skepticism score.

Appendix D: Ethics and NSD Application

Project title

Phishing Between Loneliness and Skepticism: Human Vulnerability to Social Engineering in Norwegian Higher Education

Data collected

- Anonymous survey responses (quantitative + optional open text)
- Semi-structured interviews (audio recorded with consent, pseudonymised)
- No collection of real credentials, bank details, or live phishing deployment

Legal basis and consent

Participation is voluntary. Informed consent form explains purpose, data use, storage period, right to withdraw (including deletion of interview data before analysis lock), and contact details. Participants must be 18+.

Sensitive topics

Questions about loneliness, fraud victimisation, and shame may cause discomfort. Information sheet will include links to student welfare (SiO/Samskipnaden), Mental Helses hjelpetelefon, and NSM reporting channels. No question requires disclosure of financial loss; "prefer not to say" options included.

Risk assessment

- **Participant risk:** Low. No deceptive live attacks. Scenario vignettes are clearly fictional.
- **Privacy risk:** Medium-low. Interview quotes require careful anonymisation (remove programme, names, dates).
- **Benefit:** Improved understanding of student phishing vulnerability; recommendations for HE security culture.

NSD checklist (pre-submission)

1. Finalise survey instrument and consent forms
2. Register project in NSD/REK portal
3. Obtain OsloMet faculty ethics approval if required
4. Document data processor agreements for survey platform

Appendix E: Timeline (proposed)

Table 7. Proposed research timeline aligned with ACIT long thesis programme.

Phase	Period	Activities	Deliverable
1. Alignment	Sep-Oct 2026	Supervisor meetings, refine RQ and hypotheses, literature in Zotero	Approved problem statement
2. Instrument design	Nov-Dec 2026	Validate scales, translate/adapt items, build survey	Survey v1.0 + interview guide
3. Ethics	Jan 2027	NSD application, consent forms, pilot ethics check	NSD approval
4. Pilot	Feb 2027	10-15 respondents, revise instruments	Survey v1.1
5. Main survey	Mar-Apr 2027	Recruit 120-180 via OsloMet channels	Quantitative dataset
6. Interviews	Apr-May 2027	12-18 interviews, transcription	Qualitative corpus
7. Analysis	Jun-Jul 2027	Stats + thematic analysis, integration	Results chapter
8. Writing	Aug-Nov 2027	Full thesis, supervisor feedback	Final thesis
9. Submission	Dec 2027	Final edit, formatting, submission	Master thesis

References

- Agrawal, D., Singh, K., & Sharma, D. (2025). *Understanding Gen Z's vulnerability to romance scams*. AMCIS 2025 Proceedings (Paper 36). https://aisel.aisnet.org/amcis2025/sig_sec/sig_sec/36/
- Alves, L. M., & Wilson, S. R. (2008). The effects of loneliness on telemarketing fraud vulnerability among older adults. *Journal of Elder Abuse & Neglect*, 20(1), 63-85.
- Bayl-Smith, P., Taib, R., Yu, K., & Wiggins, M. W. (2024). Response to a phishing attack: Persuasion and protection motivation in an organizational context. *Information and Computer Security*.
- Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3(2), 77-101. <https://doi.org/10.1191/1478088706qp063oa>
- Cacioppo, J. T., & Patrick, W. (2008). *Loneliness: Human nature and the need for social connection*. W. W. Norton & Company.
- Chen, Y., Li, S., Wang, Y., & Zhang, L. (2024). The effects of group discussion and role-playing training on self-efficacy, support-seeking, and reporting phishing emails. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3613904.3641943>
- Conteh, N. Y., & Schmick, P. J. (2020). Cybersecurity risks, vulnerabilities, and countermeasures to prevent social engineering attacks. In *Cyber Security and Threats: Concepts, Methodologies, Tools, and Applications* (pp. 1-20). IGI Global. <https://doi.org/10.4018/978-1-7998-6504-9.ch002>
- Cybersecurity and Infrastructure Security Agency. (n.d.). *Avoiding social engineering and phishing attacks*. Retrieved August 27, 2026, from <https://www.cisa.gov/news-events/news/avoiding-social-engineering-and-phishing-attacks>
- Du, X., Liang, J., & Chen, X. (2025). The relationship between cognitive failure and gullibility: Loneliness and impulsivity as chain mediators. *Humanities and Social Sciences Communications*, 12, Article 1435. <https://doi.org/10.1057/s41599-025-04855-3>
- European Union Agency for Cybersecurity. (2024). *ENISA threat landscape 2024*. <https://www.enisa.europa.eu/publications/enisa-threat-landscape-2024>
- Finans Norge. (2026, August). *Self-defence against fraud* (Selvforsvar mot svindel). Retrieved August 27, 2026, from <https://www.svindel.no/>
- Floyd, D. L., Prentice-Dunn, S., & Rogers, R. W. (2000). A meta-analysis of research on protection motivation theory. *Journal of Applied Social Psychology*, 30(2), 407-429. <https://doi.org/10.1111/j.1559-1816.2000.tb02323.x>
- Hadnagy, C. (2018). *Social engineering: The science of human hacking* (2nd ed.). Wiley.
- Halcyon. (2026). *AI and ransomware threat report*. Halcyon.
- Khrono & Norwegian Student Organisation. (2024, November). *7 out of 10 students are lonely*. Khrono. Retrieved August 27, 2026, from <https://www.khrono.no/7-av-10-studente-er-ensomme-snakk-med->

- Klütsch, J., Schwab, J., Böffel, C., Zimmermann, V., & Schlittmeier, S. J. (2024). Friend or phisher: How known senders and fear of missing out affect young adults' phishing susceptibility on social media. *Humanities and Social Sciences Communications*, 11(1), Article 1. <https://doi.org/10.1057/s41599-024-03412-8>
- Microsoft. (2024). *Microsoft digital defense report*. Retrieved August 27, 2026, from <https://www.microsoft.com/en-us/security/security-insider/microsoft-digital-defense-report>
- Nettvett.no & Norwegian National Security Authority. (n.d.). *Phishing and online fraud guidance* (Nettvett). Retrieved August 27, 2026, from <https://www.nettvett.no/svindel/phishing/>
- Norwegian Directorate of Health (Helsedirektoratet). (2024). *Loneliness* (Ensomhet). Retrieved August 27, 2026, from <https://www.helsedirektoratet.no/forebygging-diagnose-og-behandling/forebygging-og-levevaner/livskvalitet-trivsel-og-folkehelsearbeid/ensomhet>
- Norwegian National Security Authority (NSM). (2026). *Risk report 2026* (Risiko 2026). Retrieved August 27, 2026, from <https://nsm.no/>
- OsloMet. (2024, January 8). *Lunch Friend project* (Lunsjvenn). Den gode studentopplevelsen. Retrieved August 27, 2026, from <https://uni.oslomet.no/dengodestudentopplevelsen/2024/01/08/lunsjvenn/>
- Parker, H. J., & Flowerday, S. V. (2020). Contributing factors to increased susceptibility to social media phishing attacks. *South African Journal of Information Management*, 22(1), Article a1176. <https://doi.org/10.4102/sajim.v22i1.1176>
- Przybylski, A. K., Murayama, K., DeHaan, C. R., & Gladwell, V. (2013). Motivational, emotional, and behavioral correlates of fear of missing out. *Computers in Human Behavior*, 29(4), 1841-1848. <https://doi.org/10.1016/j.chb.2013.02.014>
- Rogers, R. W. (1975). A protection motivation theory of fear appeals and attitude change. *The Journal of Psychology*, 91(1), 93-114. <https://doi.org/10.1080/00223980.1975.9915803>
- Samskipnaden. (2025). *Feeling lonely? You are not alone* (Føler du deg ensom? Du er ikke alene!). Retrieved August 27, 2026, from <https://samskipnaden.no/en/aktuelt/fole-du-deg-ensom-du-er-ikke-alene>
- SHoT Study. (2022). *Student Health and Well-being Survey* (Studentenes helse- og trivselsundersøkelse). Retrieved August 27, 2026, from <https://shot.no/>
- Statistics Norway (SSB). (2025, December). How many people feel lonely in Norway? *Quality of Life Survey 2025* (Livskvalitetsundersøkelsen 2025). Retrieved August 27, 2026, from <https://www.ssb.no/sosiale-forhold-og-kriminalitet/levekår/statistikk/livskvalitet/artikler-om-livskvalitet/hvor-mange-fole-seg-ensomme-i-norge>
- Studentsamskipnaden i Oslo (SiO). (2024). *Loneliness among students* (Ensomhet). Retrieved August 27, 2026, from <https://www.sit.no/en/health/loneliness>

- Tjostheim, I., & Waterworth, J. A. (2020). Predicting personal susceptibility to phishing. In *Information Technology and Systems (ICITS 2020)* (pp. 540-550). Springer. https://doi.org/10.1007/978-3-030-40690-5_54
- Tjostheim, I., & Waterworth, J. A. (2022). Phishing, data-disclosure and the cognitive reflection test. In *Proceedings of the 55th Hawaii International Conference on System Sciences* (pp. 7586-7595). <https://doi.org/10.24251/HICSS.2022.926>
- Tsai, H. Y., Huang, Y. C., Lin, C. H., & Weng, S. J. (2021). Response to a phishing attack: Persuasion and protection motivation in an organizational context. *Information and Computer Security*.
- Wen, J., Yang, H., Zhang, Q., & Shao, J. (2022). Understanding the mechanisms underlying the effects of loneliness on vulnerability to fraud among older adults. *Journal of Elder Abuse & Neglect*, 34(1), 1-19. <https://doi.org/10.1080/08946566.2021.2024105>
- Wen, J., Yang, H., Zhang, Q., & Shao, J. (2024). Loneliness could lead to risk of fraud victimization for middle-aged and older adults. *Journal of Elder Abuse & Neglect*. <https://doi.org/10.1080/08946566.2024.2404040>
- WIRED. (2024). The loneliness epidemic is a security crisis. *WIRED*. Retrieved August 27, 2026, from <https://www.wired.com/story/loneliness-epidemic-romance-scams-security-crisis/>
- World Health Organization. (2023, November 3). WHO launches commission on social connection. Retrieved August 27, 2026, from <https://www.who.int/news/item/03-11-2023-who-launches-commission-on-social-connection>

Appendix F: Informed Consent Form

Project title: Phishing Between Loneliness and Skepticism: Human Vulnerability to Social Engineering in Norwegian Higher Education

Researcher: Karina Sætersdal Nilssen, Master's Programme in Applied Computer and Information Technology, OsloMet

Supervisor: [Name, faculty/department]

Contact: [student email] · [supervisor email]

Purpose of the study

You are invited to take part in research on phishing and social engineering. The study examines how loneliness, skepticism, and situational factors may relate to responses to suspicious messages. Findings will inform cybersecurity awareness and culture in higher education.

What participation involves

- **Survey (optional):** An anonymous online questionnaire (approx. 10-12 minutes) with background questions, validated loneliness items, skepticism items, and six fictional phishing scenarios. You may skip any question.
- **Interview (optional, separate consent):** If you volunteer and are selected, a 45-60 minute audio-recorded interview about experiences with suspicious messages. You will receive the interview information sheet separately.

No live phishing attacks will be used. Scenario texts are clearly fictional and contain no real links or credential requests.

Voluntary participation and withdrawal

Participation is voluntary. You may withdraw at any time without giving a reason and without negative consequences. For the anonymous survey, withdrawal means stopping before submission. For interviews, you may request deletion of your recording and transcript before analysis is locked (deadline to be stated at interview booking).

Risks and discomfort

Questions about loneliness, fraud experiences, or shame may cause mild discomfort. You may choose "prefer not to say" where offered. If you need support, contact details are provided in the information sheet (Appendix G).

Benefits

There is no direct personal benefit. Results may contribute to better security education and support for students and staff in Norwegian higher education.

Confidentiality and data storage

- Survey responses are anonymous unless you choose to leave contact details for interview follow-up (stored separately).
- Interview recordings will be pseudonymised; names, programmes, and identifying details removed from transcripts.
- Data stored on OsloMet-approved encrypted storage; access limited to researcher and supervisor.
- Data retained per NSD approval, then deleted or anonymised for long-term archive as required.

Publication

Results may be published in the master's thesis and, with further anonymisation, in academic or professional outlets. You will not be identifiable in any publication.

Legal basis

Processing is based on your consent (GDPR Art. 6(1)(a) and, where applicable, Art. 9(2)(a) for sensitive topics). The project is registered with NSD / REK before data collection begins.

Consent statement

By ticking the box below (or signing below for paper use), I confirm that:

- I am 18 years or older.
- I have read and understood this consent form and the information sheet (Appendix G).
- I agree to participate in the **survey** described above.
- I understand I may withdraw at any time.

☐ I consent to participate in the survey.

☐ I do **not** wish to participate.

[Optional for interview recruitment] ☐ I am willing to be contacted about a follow-up interview (contact details collected on a separate form).

Digital implementation: consent captured via survey platform checkbox with timestamp. Paper copy available on request.

Appendix G: Participant Information Sheet

Welcome - thank you for considering participation in this master's thesis research at OsloMet.

What is this study about?

Phishing attacks often succeed not because people lack knowledge, but because messages imitate things people need or fear losing: belonging, safety, order, or help. This study explores how loneliness, general skepticism, and situational pressure may relate to responses to suspicious digital messages among adults in higher education in Norway.

Who can participate?

Adults aged 18+ who study or work in Norwegian higher education and use digital communication regularly.

What will I be asked to do?

1. **Online survey:** Answer questions about background, loneliness (standard short scale), online skepticism, six short fictional phishing scenarios, and security habits. Estimated time: 10-12 minutes.
2. **Optional interview:** Some survey respondents may be invited to a confidential interview (45-60 min, video or in person). Separate consent applies.

Important safeguards

- No real phishing links or password collection.
- Survey is anonymous unless you separately volunteer contact details for interviews.
- You may skip questions or stop at any time.
- Interview quotes will be anonymised before use in the thesis.

Possible discomfort

Some questions ask about loneliness or whether you have received scam messages. You may find this mildly uncomfortable. You do not have to share details of any financial loss. "Prefer not to say" is always an option.

Support resources

Service	Contact
OsloMet student welfare (SiO)	https://sio.no/en
Mental Health helpline (Mental Helse)	116 123 (24/7, Norway)

Service	Contact
Report digital fraud / phishing	NSM: https://nsm.no · Finans Norge: https://www.svindel.no
OsloMet IT security	[OsloMet IT helpdesk URL / email]

What happens to my data?

Data are processed by Karina Sætersdal Nilssen under supervisor oversight. Survey data stored on an approved platform; interview recordings on encrypted OsloMet storage. Only the research team has access. Data are kept only as long as NSD approval requires, then deleted or permanently anonymised.

Your rights

You have the right to access, correct, or request deletion of personal data (where not already anonymous), restrict processing, and lodge a complaint with Datatilsynet (Norwegian Data Protection Authority). Contact the researcher or supervisor for questions about your rights.

Questions?

Researcher: Karina Sætersdal Nilssen · [email]

Supervisor: [name] · [email]

OsloMet data protection: [privacy contact if applicable]

Version 1.0 · August 2026 · To be updated after NSD approval